

Lesson 10

Managing Data Security

Topic 10A

Understand Data Security Concepts

Exam Objectives Covered

- 3.1 Summarize data security concepts

The Business Impact of Data Security

- Data value prioritization
- Lifecycle management
- Data management policy
- The cost of security versus risk and/or replacement

Data Encryption

- Goals of encryption
 - Confidentiality
 - Integrity

```
[root@localhost ~]# md5sum file1.txt
d41d8cd98f00b204e9800998ecf8427e  file1.txt
[root@localhost ~]#
[root@localhost ~]# █
```

Generate a checksum by using the md5sum command.

- Non-repudiation
- Data in transit versus data at rest

Retention Policies

- Data retention
- Physical storage
 - On-site
 - Off-site

Additional Data Security Measures

- UEFI/BIOS passwords
- Boot loader passwords

Guidelines for Understanding Data Security

- Encryption goals: confidentiality, integrity, non-repudiation
- Balance data security cost and risk
- Retention policies
- Data in transit may be protected by various protocols
- Data at rest may be protected by drive and file encryption
- Tools can detect changes to data
- On-site versus off-site or cloud storage
- UEFI/BIOS and boot loader passwords give additional authentication layer

Review Activity: Data Security Concepts



Review Activity

Topic 10B

Manage Data Security

Exam Objectives Covered

- 3.4 Explain data security risks and mitigation strategies

Risks and Mitigation

- Hardware failure
- Malware
- Data corruption
- Insider threats
- Data Loss Prevention (DLP)
- Unwanted:
 - Duplication (theft)
 - Publication (theft)
 - Access (backdoor, social engineering)
- Breach

Industry and Government Regulations

- PII
- PCI DSS
- HIPPA

Breaches

- PII numbers
 - Identification such as SSN
 - Bank account
 - Passport
 - Driver's license
 - Bank card/credit card
 - Telephone
 - Birth date
- Other
 - Name
 - Address
 - Gender

Guidelines for Managing Data Security

- Implement strong SIEM
- Role separation and rotation
- Log file management and analysis
- Stay aware of regulations and requirements
- Use best practices
- Stay compliant with data retention requirements

Review Activity: Data Security



Review Activity

Topic 10C

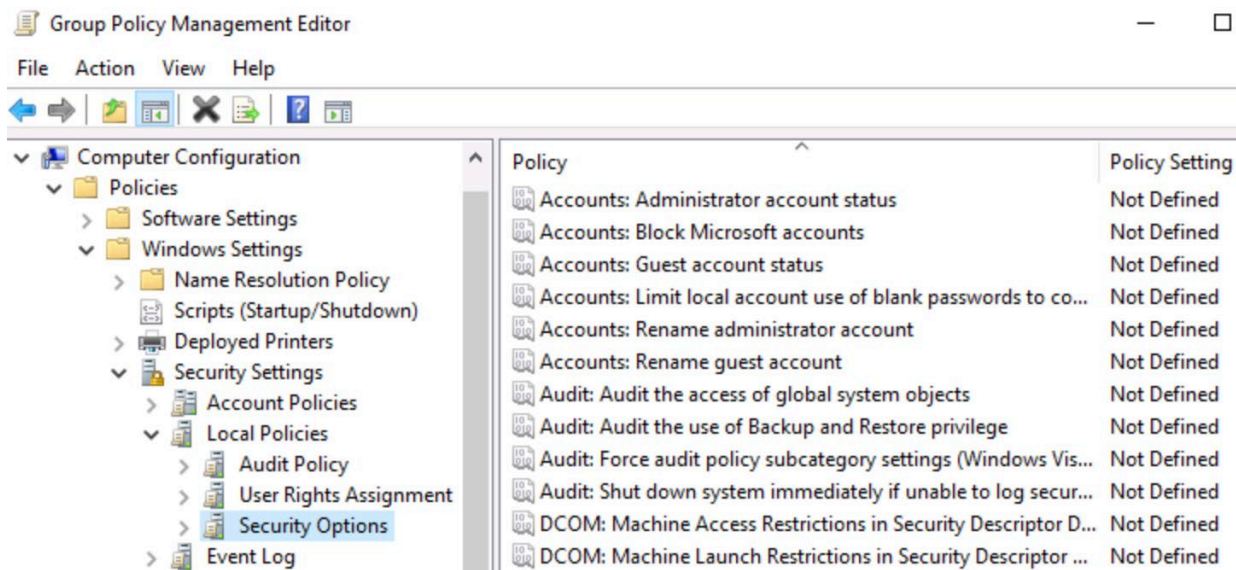
Troubleshoot Data Security

Exam Objectives Covered

- 4.6 Given a scenario, troubleshoot security problems

General Troubleshooting

- Resource access
- Group policy configurations



AD Group Policy Security Options. (Screenshot courtesy of Microsoft.)

General Troubleshooting (slide 2)

- Firewall configurations
 - Too restrictive
 - Too permissive

AD Group Policy Security Options. (Screenshot courtesy of Microsoft.)

General Troubleshooting (slide 3)

- Service configurations

Systemctl start sshd

```
top - 15:11:45 up 22 min, 1 user, load average: 0.00, 0.02, 0.07
Tasks: 216 total, 1 running, 215 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.7 us, 0.0 sy, 0.0 ni, 98.3 id, 0.3 wa, 0.7 hi, 0.0 si, 0.0 st
MiB Mem : 3927.9 total, 1917.2 free, 776.6 used, 1234.1 buff/cache
MiB Swap: 1963.0 total, 1963.0 free, 0.0 used. 2906.3 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1691	damon	20	0	3835420	255780	122460	S	0.7	6.4	0:05.34	gnome-shell
2977	root	20	0	234964	4604	3960	R	0.3	0.1	0:00.02	top
1	root	20	0	174992	17428	11556	S	0.0	0.4	0:01.27	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kthreadd
3	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	rcu_gp
.	.	.	.	.	.	.	.	.	.	.	.

Managing processes by using top.

General Troubleshooting (slide 4)

- Common concerns
 - File integrity
 - Improper privilege escalation (excessive access)
 - Applications won't load
 - Cannot access network shares
 - Unable to open files

Troubleshooting Tools and Their Uses

- Port scanners
- Packet sniffers
- Telnet clients
- Anti-malware
- Antivirus
- File integrity
- User access controls
- Service management tools
- Group policy

Review Activity: Data Security Troubleshooting



Review Activity

Lesson 10

Summary

